

Based on your latest architecture, research documents, and implementation direction, here's the complete understanding of **SentinelAI (AI-NEXUS)**.

SentinelAI (AI-NEXUS)

Enterprise AI-Powered Cyber Risk Intelligence Platform

One-Line Definition

SentinelAI is an **AI-powered Cyber Risk Intelligence Platform** that collects security events from multiple enterprise security tools, correlates them into incidents, evaluates business-aware cyber risk using deterministic analytics, and finally generates explainable AI recommendations backed by evidence.

It is **not another SIEM**.

It is an **Intelligence Layer** that sits above existing security tools.

The Problem Statement

Current Enterprise Problem

Large organizations already have many cybersecurity products:

- SIEM (Wazuh, Splunk, QRadar)
- EDR (CrowdStrike, Microsoft Defender)
- IAM (Okta, Azure AD)
- Vulnerability Scanners (OpenVAS, Nessus)
- Firewalls
- Cloud Security (AWS Security Hub, Azure Defender)
- Compliance Platforms

Each tool performs its own job well.

The problem is **they operate independently**.

Example

An attacker performs an SSH brute-force attack.

What happens?

Wazuh

SSH Brute Force Detected

Suricata

Port Scan Detected

OpenVAS

Apache has CVE-2024-XXXX

IAM

John logged in successfully

Firewall

Connection Accepted

Every tool produces

its own

alert.

Now imagine

5000 alerts/day.

The SOC analyst must manually ask

Are these alerts related?

Is this one attack?

Is this dangerous?

Which system is affected?

What should I fix first?

This takes hours.

Existing Problems

1. Alert Fatigue

SOC teams receive thousands of alerts every day.

Most are

- duplicates
- isolated
- low priority

Analysts spend more time filtering alerts than investigating attacks.

2. No Context

Current tools tell you

What happened

They don't tell you

Why it matters.

Example

SSH Login Success

Was it

normal?

after brute force?

administrator?

critical server?

Nobody knows.

3. No Relationship Between Events

Today's tools think

Port Scan

SSH Login

Privilege Escalation

OpenVAS CVE

are

four separate alerts.

Reality

They're one attack.

4. CVSS is Not Enough

OpenVAS says

CVSS 9.8

But

Server is offline

↓

No business impact

Risk is actually low.

Another vulnerability

CVSS 6.5

↓

Payroll Database

↓

Finance Server

↓

Internet Facing

Risk is much higher.

Traditional tools cannot understand business context.

5. AI Hallucination

Many AI security tools simply feed logs into an LLM.

LLMs hallucinate.

No evidence.

No traceability.

Not suitable for enterprise security.

SentinelAI Solution

Instead of replacing security tools,

SentinelAI connects all of them.

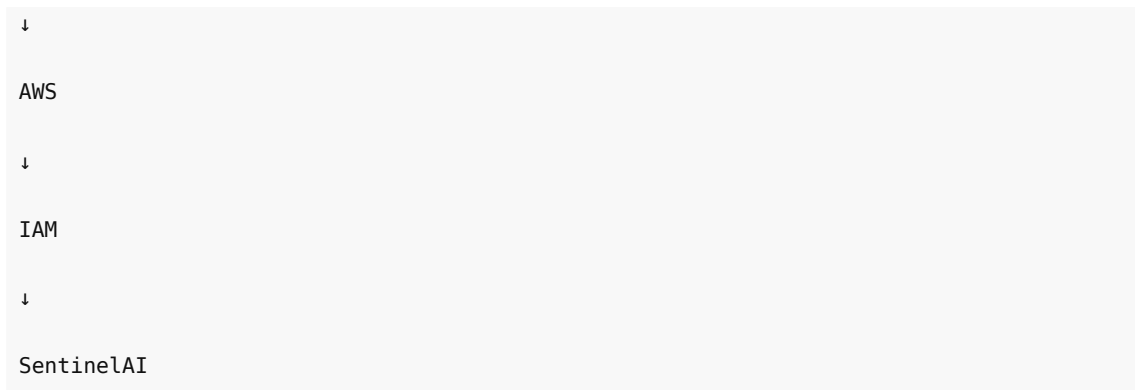
Wazuh

↓

Suricata

↓

OpenVAS

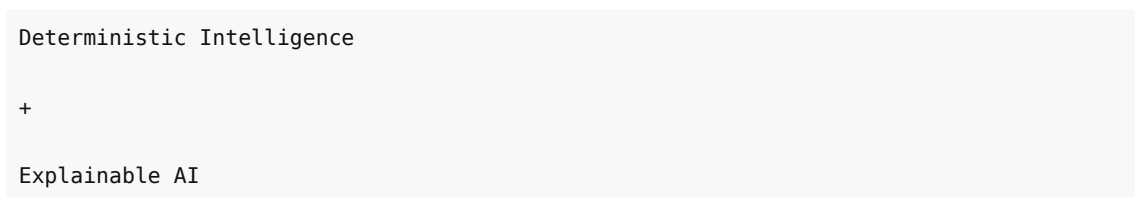


SentinelAI

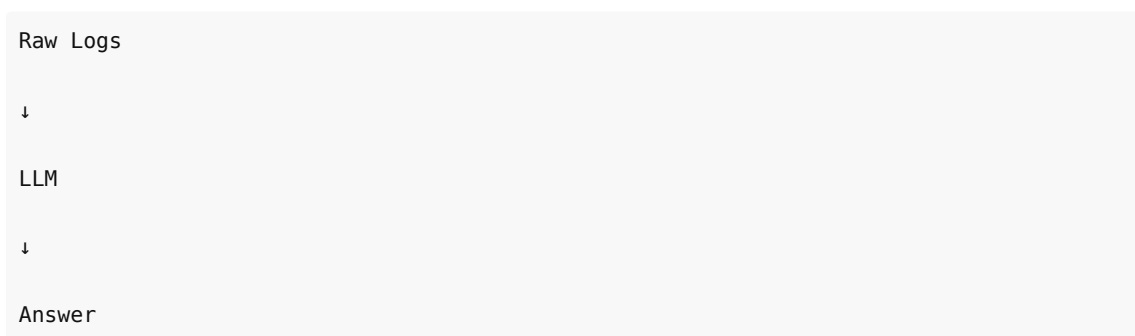
- correlates events
- understands relationships
- calculates business-aware risk
- uses AI only after deterministic analysis
- provides explainable recommendations

Core Innovation

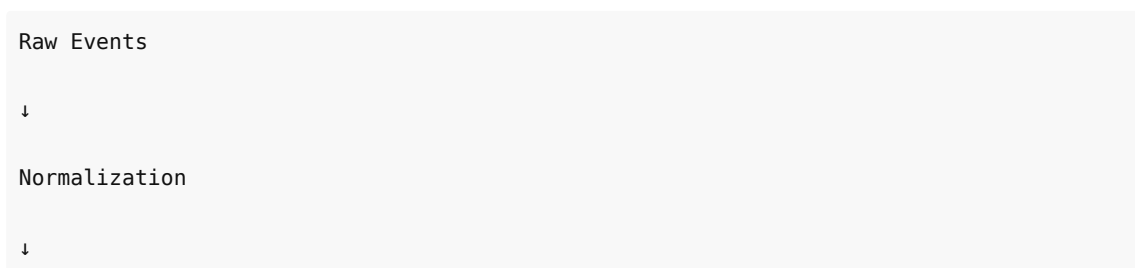
The biggest innovation is



instead of



SentinelAI does



Correlation

↓

Knowledge Graph

↓

Risk Engine

↓

AI

↓

Verification

↓

Dashboard

AI never decides risk.

AI only explains verified facts.

This is the core research principle of SentinelAI.

Complete Pipeline

Stage 1

Enterprise Data Ingestion Engine

Purpose

Collect security information from different tools.

Examples

- Wazuh
- Suricata
- OpenVAS
- AWS
- IAM

Responsibilities

- Collect events
- Authenticate

- Validate
- Remove duplicates
- Publish events

Output

```
Security Events
```

Stage 2

Data Normalization Engine

Every tool has different formats.

Example

Wazuh

```
severity:12
```

OpenVAS

```
risk=critical
```

AWS

```
HIGH
```

Normalization converts everything into

one enterprise format.

Output

```
Canonical Security Entity
```

Stage 3

Threat Correlation Engine + Knowledge Graph

Instead of

```
Alert
```

```
Alert
```

```
Alert
```

build

Incident

└ Port Scan

└ Hydra

└ Login

└ Privilege Escalation

└ Vulnerability

Also

create

graph

User

↓

Machine

↓

Application

↓

Database

↓

Vulnerability

Everything becomes connected.

Output

Incident Graph

Stage 4

Risk Intelligence Engine

The brain of SentinelAI.

Uses

- Attack Path

- Blast Radius
- Asset Criticality
- Compliance
- Business Impact
- Graph Analytics
- MITRE Mapping

Produces

Risk Score

Priority

Evidence

Confidence

Business Impact

This stage **does not use AI**. It performs deterministic graph analysis and generates structured risk evidence for the next stage.

Stage 5

AI Decision Intelligence Engine

Now AI starts.

Input

Verified Risk Evidence

AI generates

- Executive Summary
- Technical Summary
- Root Cause
- Remediation
- Business Explanation
- Analyst Guidance

AI never receives raw logs.

Only verified evidence.

Stage 6

Explainability Engine

This stage verifies AI.

Checks

Does this recommendation exist in graph?

Is evidence attached?

Does risk score match?

Are claims supported?

Then

attaches

- citations
- confidence
- graph references
- supporting evidence

Output

Verified Response

Stage 7

API Gateway

Provides secure access.

Responsibilities

- JWT
- RBAC
- Rate Limiting
- Validation
- Routing
- Logging
- API Versioning

Everything enters through Stage 7 before reaching users.

Stage 8

Incident Dashboard & AI Assistant

Different users receive different views.

Executive

- Enterprise Risk
- KPIs

- Compliance
- Trends

SOC Analyst

- Incident Timeline
- MITRE Techniques
- Attack Path
- Vulnerabilities
- Risk Score

AI Assistant

Questions

Why is Payroll Server critical?

↓

Show attack path

↓

Recommend remediation

↓

Explain business impact

The dashboard consumes validated responses from the API Gateway and presents role-specific experiences such as executive dashboards, SOC investigations, graph visualization, and an AI assistant.

Runtime Example

Suppose an attacker launches an attack.

Hydra SSH Brute Force

↓

Suricata detects scan

↓

Wazuh detects brute force

↓

SSH Login Success

↓

Privilege Escalation

↓

OpenVAS finds Apache CVE

↓

Stage 1 collects events

↓

Stage 2 normalizes data

↓

Stage 3 correlates everything into one incident

↓

Stage 4 calculates risk

↓

Critical

Risk Score 9.7

MITRE Techniques

Business Impact

↓

Stage 5 generates executive + analyst explanation

↓

Stage 6 validates every claim and attaches evidence

↓

Stage 7 exposes secure APIs

↓

Stage 8 displays the incident dashboard

Instead of showing five unrelated alerts, SentinelAI presents **one verified incident with its timeline, risk score, evidence, MITRE mapping, AI explanation, and recommended remediation.**

Why SentinelAI Stands Out

Most security products answer:

"What happened?"

SentinelAI answers:

- **What happened?**
- **Why did it happen?**
- **Which business assets are affected?**
- **How severe is it in organizational context?**
- **What evidence supports this conclusion?**
- **What should the SOC analyst or executive do next?**

That shift—from isolated alerts to **explainable cyber risk intelligence**—is the fundamental vision and differentiator of your project.

Based on your **latest research**, **current implementation**, and the refined architecture (Threat Correlation Engine, Risk Intelligence Engine, AI Decision Engine, Explainability Engine, API Gateway, Dashboard), the best distribution is **module ownership**, not stage ownership. This minimizes merge conflicts while preserving the 8-stage architecture.

SentinelAI Team Distribution (4 Members)

Member 1 — Security Data Acquisition & Simulation

Owns: Stage 1 (Enterprise Data Ingestion Engine)

Responsibilities

- Build the complete attack simulation environment.
- Integrate all security data sources.
- Deliver validated security events to downstream modules.

Modules

```
/simulation/  
/connectors/  
/ingestion/  
/config/  
/docker/  
/security_sources/
```

Tasks

Attack Simulation

- Configure Attacker VM
- Configure Target VM
- Hydra attack simulation
- Nmap scan simulation

- Successful SSH login simulation
- Privilege escalation simulation
- Vulnerability simulation

Security Tools

- Wazuh
- Suricata
- OpenVAS
- Optional AWS Connector
- Mock AWS findings
- XML parser

Ingestion Engine

- Connector Manager
- Scheduler
- Authentication Manager
- Payload Validator
- Duplicate Detection
- Metadata Enrichment
- Event Publisher

Output

Produces only

Security Events (JSON)

No normalization.

No AI.

No dashboard.

Member 2 — Intelligence Core

Owns

Stage 2

Stage 3

Stage 4

This is the backend intelligence engineer.

Modules

```
/normalization/  
/graph/  
/risk/  
/models/  
/neo4j/  
/analytics/
```

Stage 2

Data Normalization Engine

Implement

- Vendor Mapping
- Schema Mapping
- Canonical Entity Model
- Timestamp Standardization
- Severity Mapping
- Entity Resolution
- Data Validation
- Metadata Enrichment

Output

Canonical Entities

Stage 3

Threat Correlation Engine + Knowledge Graph Builder

Implement

- Incident Correlation
- Timeline Correlation
- MITRE Mapping
- Neo4j Nodes
- Neo4j Relationships
- Graph Builder
- Graph Updates

Build

Incident

- └ Scan
- └ Hydra
- └ Login
- └ Priv Esc
- └ Vulnerability

instead of individual alerts.

Stage 4

Risk Intelligence Engine

Implement

- Attack Path Discovery
- Blast Radius
- Business Impact
- Asset Criticality
- Compliance Mapping
- Contextual Risk Score
- Risk Priority
- Confidence
- Risk Evidence Payload

Output

```
RiskEvidence
```

No AI generation.

Member 3 — AI Intelligence Layer

Owns

Stage 5

Stage 6

This member owns every LLM-related component.

Modules

```
/ai/  
/langgraph/  
/prompts/  
/explainability/  
/verification/  
/rag/
```

Stage 5

AI Decision Intelligence

Implement

- LangGraph workflow
- Prompt Builder
- Evidence Retrieval
- Recommendation Generator
- Executive Summary
- Technical Summary
- Remediation Generator
- Incident Explanation
- Business Explanation

Input

RiskEvidence

Output

AIResponse

Stage 6

Explainability Engine

Implement

- Evidence Validator
- Graph Verification
- Rule-based Confidence Score
- Citation Generator
- Unsupported Claim Detection
- Evidence Attachment
- Final Verified Response

Output

VerifiedResponse

No frontend.

Member 4 — Platform Experience

Owns

Stage 7

Stage 8

Modules

/gateway/
/frontend/
/dashboard/
/chat/
/api/
/auth/

Stage 7

API Gateway

Implement

- FastAPI
- JWT
- RBAC

- Rate Limiting
- Request Validation
- Response Standardization
- API Versioning
- Audit Logging

Expose APIs

```
/incident  
  
/risk  
  
/dashboard  
  
/chat  
  
/graph  
  
/reports
```

Stage 8

Incident Dashboard

Implement

Executive Dashboard

- Enterprise Risk
- KPI Cards
- Compliance
- Trends

SOC Dashboard

- Incident Timeline
- MITRE Flow
- Attack Path
- Vulnerability Panel
- Risk Score

Knowledge Graph

- React Flow
- Cytoscape

AI Chat

- Chat Interface
- Streaming Responses

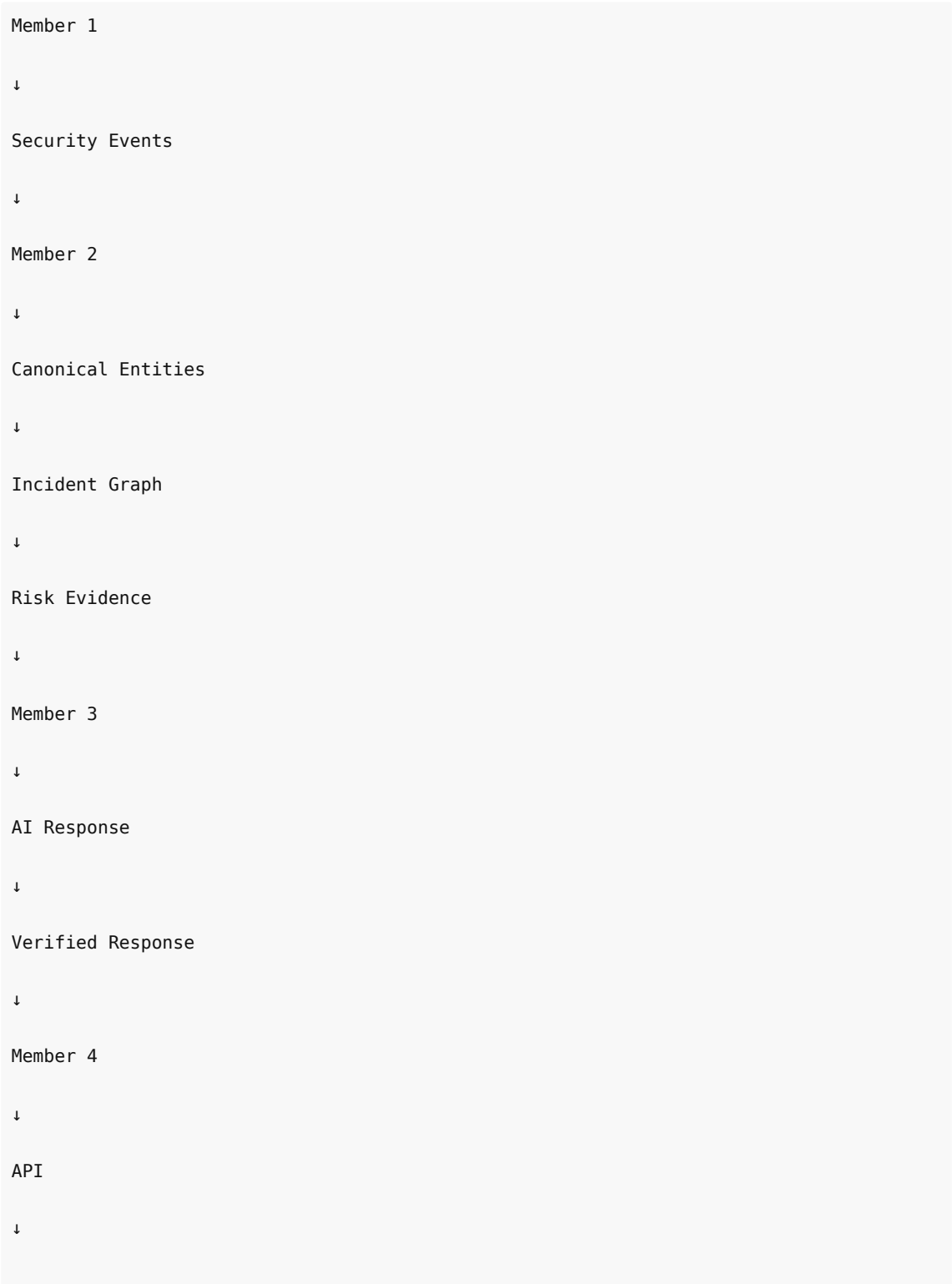
Reports

- PDF
- CSV

Notifications

- Incident Alerts

Integration Order



Merge Conflict Prevention

Member	Owns	Never Touches
Member 1	/simulation, /connectors, /ingestion, /config	Graph, AI, Frontend
Member 2	/normalization, /graph, /risk, /neo4j	Connectors, AI, Frontend
Member 3	/ai, /langgraph, /prompts, /explainability	Graph internals, Frontend
Member 4	/gateway, /frontend, /dashboard, /chat	Connectors, Graph, AI internals

Shared Interface Contracts (decide once, then freeze)

Only these artifacts are shared across members:

- 1. **SecurityEvent** (Member 1 → Member 2)
- 2. **CanonicalEntity** (Stage 2 → Stage 3)
- 3. **RiskEvidence** (Member 2 → Member 3)
- 4. **VerifiedResponse** (Member 3 → Member 4)

By agreeing on these JSON schemas at the start, each member can develop independently with minimal merge conflicts. This distribution also aligns with your updated research: Member 1 focuses on data acquisition and simulation, Member 2 builds the deterministic intelligence core (normalization, correlation, graph, and risk), Member 3 implements the AI and explainability pipeline, and Member 4 delivers the secure APIs and user-facing experience.